

Gold Star

Lab 1: Descriptive Paper

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Version 3

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1 Introduction

Millions of people struggle with ADHD symptoms such as starting tasks, staying focused, or finishing what they have started (CDC, 2024). While suffering from these symptoms, building and maintaining routines is an effective solution, but often hard to achieve. Routines often last anywhere from three days to “part of a month” according to ADHD Philadelphia.

People who struggle with routine management need a solution that is built with them in mind, allowing them the grace and flexibility needed to be successful. Gold Star is a web-based habit-building and time management platform designed specifically for those with ADHD or suffering from ADHD-like symptoms. It introduces the user to flexible scheduling, giving them the ability to change their routine when they need it, or are not feeling the most productive.

2 Gold Star Product Description

Gold Star is a web-based time management platform that allows those who struggle with ADHD symptoms to manage their time, tasks, and daily responsibilities. Users can plan, track, and maintain routines with the flexibility that comes with adaptive scheduling. Gold Star’s Adaptive Scheduling algorithm allows schedules to evolve with the user’s needs, modifying schedules based on the user’s productivity levels and mood. With scheduling that incorporates researched productivity boosting techniques and habit-building strategies, Gold Star helps users stay consistent, complete tasks, and better manage their daily lives.

2.1 Key Product Features and Capabilities

Gold Star’s main user interface consists of a calendar view. This allows the user to clearly see their curated schedule with the ability to add tasks to it or change their current schedule if

they deem it necessary. This interface also allows the prioritization and grouping of tasks, a feature that gives the user the ability to focus on the tasks most important to them while arranging similar tasks to be scheduled together. This can reduce activity switching, a cognitive process that may cause a decline in productivity (American Psychological Association 2006).

Gold Star's unique Adaptive Scheduling system creates a distinctive schedule carefully crafted for each individual user, allowing changes to priorities and rescheduling missed tasks when needed. Gold Star evolves with the user, allowing changes in productivity level. This lets them feel successful even when they are not necessarily feeling especially productive. Instead of punishing the user for missing goals, Gold Star features a "Roll Over" system. This allows the user to move their missed tasks to another day, giving them the flexibility they need without any added stress.

All these features come together to help users overcome common ADHD challenges such as task initiation, consistency, commitment, and completion. In this way, Gold Star can assist in improving productivity and reducing stress.

2.2 Major Components (Hardware/Software)

The system requires no specialized hardware. Users can access the application through any off-shelf device with a modern web browser. The server side runs on a single hosting environment supporting the Node.js web server and PostgreSQL database, with deployments being automatically pushed through GitHub. As shown in Figure 1, the system follows a three-tier structure: a Presentation Layer (the web-based UI), an Application Layer (the Node.js web server / API with authentication, and a notification handler), and a Data Layer (PostgreSQL database storing the encrypted user data, tasks, and routine data).

The software to be developed includes the UI /UX built with HTML, CSS, and JavaScript. It will feature a calendar view, routine dashboard, productivity level, and statistics dashboard. Node.js web server handles routing, logic, authentication components, and WebSocket-based notification handler. PostgreSQL database to store user account information, tasks, routine data, and sensitive user information kept in an encrypted form. Development is supported by off-the-shelf tools: Visual Studio Code, Git and GitHub, GitHub Actions for CI/CD, Jest for testing, and Javadoc for documentation.

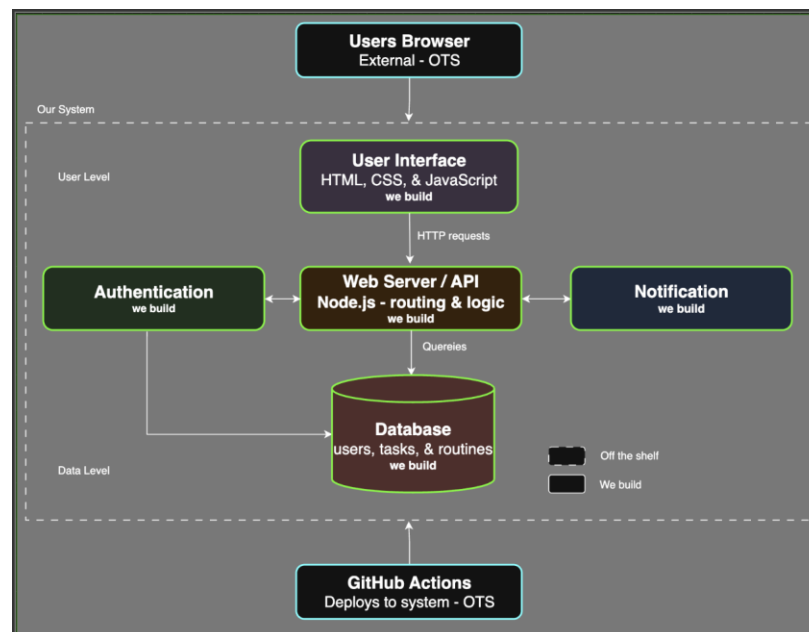


Figure 1: Major Functional Component Diagram

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3 Glossary

Activity Alternation – Switching from an activity requiring high mental focus to an activity requiring low mental focus.

Activity Switching – The cognitive process of disengaging from one task and shifting mental focus to another task.

Adaptive Scheduling – To schedule a task which can be completed anytime within a set timeframe. For example, a task which can be completed anywhere inclusively between 7:00 P.M. and 10:00 P.M.

Application Programming Interface (API) - A set of standard rules, protocols, and definitions allowing different software applications to interact with one another.

Attention-Deficit/Hyperactivity Disorder (ADHD) - A common neurodevelopmental disorder that alters the brain's ability to plan, focus, remember instructions, and manage emotions.

Cascading Style Sheets (CSS) - A style sheet language used to determine the visual presentation, layout, and aesthetics of a website.

Git - A digital version control system intended to preserve a history of saved modifications, additions, and deletions to software source code.

GitHub - A cloud-based platform where developers can collaborate on software code.

Gold Star - A web-based habit-building and time management platform designed specifically for individuals with ADHD.

Habit Stacking – Interspersing new habits with existing habits to make task completion easier without feeling overwhelmed.

Hyperactivity - A state of being unusually or abnormally active.

HyperText Markup Language (HTML) - The standard markup language used to create and structure webpages.

JavaScript (JS) - A lightweight programming language used to build dynamic interactivity on web pages.

Jest - An open-source JavaScript testing framework.

JSDoc - An API documentation generator and syntax standard for JavaScript.

Pomodoro Technique – A time-management technique which separates work into 25-minute increments and 5-minute breaks. The standard Pomodoro process repeats this technique 4 times, then a longer break between 20 to 30 minutes is taken before the process repeats.

PostgreSQL - An open-source RDBMS that stores, manages, and retrieves data.

Reinforce - To make something stronger, more effective, or more compelling by adding support.

Relational Database Management System (RDBMS) - Software used to store, manage, query,

and retrieve data structured in table form.

Rigid Scheduling – To schedule a high-priority task which must either be initiated at or completed before an established time. For example, a task which must start at 6:00 P.M. or must be completed by a deadline of 6:00 P.M.

Routine Adaption – A requirement for an individual to change one's schedule or create new routines.

Routine - A sequence of actions regularly followed, or a fixed program.

Task Deviation – When a task has an unexpected change regarding completion. A task may be accomplished out of order in one's schedule, or it may not be accomplished at all.

Visual Studio Code (VSCode) - A development environment used for software engineering and web development.

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